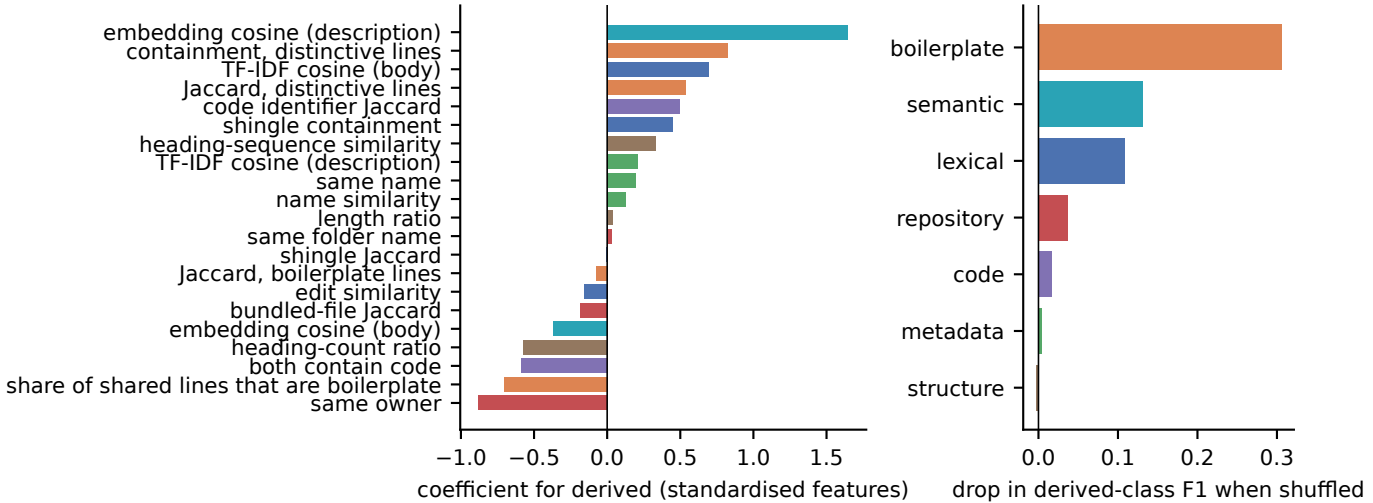
(a) logistic regression
embedding cosine (description)
containment, distinctive lines
TF-IDF cosine (body)
Jaccard, distinctive lines
code identifier Jaccard
shingle containment
heading-sequence similarity
TF-IDF cosine (description)
same name
name similarity
length ratio
same folder name
shingle Jaccard
Jaccard, boilerplate lines
edit similarity
bundled-file Jaccard
embedding cosine (body)
heading-count ratio
both contain code
share of shared lines that are boilerplate
same owner
−1.0
−0.5
0.0
0.5
1.0
1.5
coefficient for derived (standardised features)
(b) gradient boosting, held-out folds
boilerplate
semantic
lexical
repository
code
metadata
structure
0.0
0.1
0.2
0.3
drop in derived-class F1 when shuffled